

ASHEVILLE, NC, USA

WMO: 723150

Lat: 35.432N

Lon: 82.538W

Elev: 2117

StdP: 13.61

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 03812

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.9	19.4	-0.1	5.9	24.1	4.1	7.4	26.2	27.3	29.0	24.9	28.8	12.3	340	0.425

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	87.9	70.7	85.9	70.2	83.9	69.6	73.7	82.7	72.7	81.2	71.7	79.7	7.8	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.3	124.8	76.9	70.2	120.1	76.0	69.2	116.2	75.3	38.7	82.7	37.6	81.4	36.7	79.8	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	19.2	17.1		8.0	91.0	5.1	2.3	4.3	92.7	1.3	94.0	-1.6	95.3	-5.3	97.0
			DB	6.5	75.8	4.9	2.2	3.0	77.4	0.2	78.7	-2.6	79.9	-6.1	81.6
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.5	37.5	41.2	47.6	56.4	64.1	71.0	73.8	73.1	67.4	57.2	47.2	40.9
	DBStd	14.65	9.88	9.00	8.81	7.39	6.17	4.25	3.46	3.41	5.32	7.25	7.94	8.71
	HDD50	1316	401	267	151	26	1	0	0	0	0	20	144	306
	HDD65	4000	852	667	538	267	92	8	0	1	37	256	534	748
	CDD50	3702	14	20	78	219	437	630	738	717	522	244	60	23
	CDD65	912	0	0	0	10	64	188	273	252	109	15	1	0
	CDH74	6335	0	2	16	167	542	1313	1850	1613	723	105	4	0
	CDH80	1610	0	0	0	15	97	340	545	442	158	13	0	0
Wind	WSAvg	6.2	8.3	8.2	8.1	7.4	5.8	4.8	4.4	4.1	4.3	5.5	6.8	7.4
Precipitation	PrecAvg	59.0	4.9	4.6	5.4	4.7	4.3	5.6	5.5	5.6	5.2	3.8	4.2	5.3
	PrecMax	84.6	11.7	9.3	10.1	8.7	10.0	10.6	14.2	10.1	19.3	8.8	8.6	10.2
	PrecMin	39.7	1.3	1.9	2.0	2.1	1.5	1.6	1.9	1.0	0.9	0.3	1.5	2.3
	PrecStd	10.1	2.5	1.8	1.9	1.6	2.1	2.2	2.7	2.2	3.5	2.2	1.8	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.4	70.9	76.2	82.1	86.4	89.3	90.3	90.4	87.3	81.3	74.0	68.1
		MCWB	56.0	58.5	58.1	62.4	66.9	69.9	72.2	71.6	67.6	67.6	58.8	57.5
	2%	DB	61.6	64.7	71.5	78.7	82.7	86.4	87.6	87.3	84.2	76.8	69.0	63.6
		MCWB	54.7	53.6	56.1	60.7	66.0	69.6	71.6	71.0	67.8	63.3	55.9	56.4
	5%	DB	57.7	60.8	67.5	75.3	79.9	84.0	85.5	84.7	81.5	73.4	65.0	59.3
		MCWB	51.5	51.8	54.2	59.0	65.2	68.9	71.3	70.7	67.4	61.7	55.8	52.9
	10%	DB	53.3	56.5	63.2	71.6	76.8	81.4	83.3	82.2	78.5	70.1	61.7	55.4
		MCWB	48.3	48.4	52.2	57.4	63.8	68.0	70.8	69.9	66.5	60.3	54.3	49.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.9	61.1	61.3	66.5	70.6	73.5	75.5	75.1	72.3	70.6	64.4	61.9
		MCDB	63.3	67.9	69.6	77.5	80.0	83.3	85.1	83.9	80.4	77.5	67.8	65.8
	2%	WB	56.4	57.7	59.1	63.8	68.7	72.2	74.1	73.5	70.9	67.4	60.9	58.2
		MCDB	60.0	62.5	67.7	73.4	77.7	81.5	83.2	82.3	78.6	72.3	64.9	61.5
	5%	WB	52.7	53.8	57.1	61.8	67.4	71.0	73.0	72.4	69.5	64.8	58.6	54.6
		MCDB	56.4	58.1	64.3	70.2	75.8	79.9	81.6	80.6	76.5	69.7	63.1	57.7
	10%	WB	48.7	49.8	54.6	59.8	65.9	69.8	72.0	71.4	68.3	62.6	55.6	50.9
		MCDB	52.2	54.6	60.6	67.7	73.7	78.3	80.0	78.9	74.6	67.5	59.8	54.3
Mean Daily Temperature Range	5% DB	MDBR	19.1	19.9	21.3	22.9	21.4	19.6	18.4	18.4	19.1	21.3	21.8	18.5
		MCDBR	25.4	27.3	28.2	29.2	25.1	23.5	21.5	22.0	23.5	25.6	27.0	24.1
		MCWBR	17.0	17.1	15.3	13.8	11.0	9.1	8.4	8.4	9.4	12.8	15.9	16.0
	5% WB	MCDBR	20.1	21.5	23.0	22.9	20.9	20.0	19.1	18.7	18.5	18.0	19.5	19.2
		MCWBR	15.5	16.2	14.5	12.3	9.9	8.7	8.1	7.9	8.3	10.5	14.3	14.9
Clear-Sky Solar Irradiance	taub	0.294	0.303	0.333	0.373	0.426	0.452	0.493	0.490	0.386	0.352	0.313	0.292	
	taud	2.544	2.512	2.455	2.344	2.211	2.172	2.087	2.074	2.350	2.440	2.515	2.579	
	Ebn at Noon	291	300	298	289	273	265	253	252	276	277	280	285	
	Edh at Noon	25	29	33	39	46	47	51	51	37	31	26	23	
All-Sky Solar Radiation	RadAvg	822	1022	1351	1673	1839	1935	1800	1686	1460	1219	918	709	
	RadStd	51	109	86	127	137	134	125	78	131	126	80	64	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.08	N/A	N/A	N/A	N/A	N/A	N/A	-209	+301	+158
(47)	Regional (41 neighbors)	+0.97	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+277	+180

Nomenclature: See separate page